Newman-Penrose Formalism Calculations

Stephani et al., Equation (14.34)

Metric

$$ds^2 = -c^2 F(t)^{2\,a_4}\, G(t)^{2\,b_4}\, \mathrm{d}t \otimes \mathrm{d}t + c^2 F(t)^{2\,a_1}\, G(t)^{2\,b_1}\, \mathrm{d}x \otimes \mathrm{d}x + c^2 F(t)^{2\,a_2}\, G(t)^{2\,b_2}\, e^{(2\,c_2 x)} \mathrm{d}y \otimes \mathrm{d}y + c^2 F(t)^{2\,a_3}\, G(t)^{2\,b_3}\, e^{(2\,c_3 x)} \mathrm{d}z \otimes \mathrm{d}z$$

Coordinates

$(x^\mu) = (t, x, y, z)$

Null Tetrad

Covariant components

$$l = \left(\sqrt{\frac{1}{2}} c F(t)^{a_4}\, G(t)^{b_4}\right) dt + \left(\sqrt{\frac{1}{2}} c F(t)^{a_1}\, G(t)^{b_1}\right) dx$$

$$n = \left(\sqrt{\frac{1}{2}} c F(t)^{a_4}\, G(t)^{b_4}\right) dt + \left(-\sqrt{\frac{1}{2}} c F(t)^{a_1}\, G(t)^{b_1}\right) dx$$

$$m = \left(\sqrt{\frac{1}{2}} c F(t)^{a_2}\, G(t)^{b_2}\, e^{(c_2 x)}\right) dy + \left(i\sqrt{\frac{1}{2}} c F(t)^{a_3}\, G(t)^{b_3}\, e^{(c_3 x)}\right) dz$$

$$\overline{m} = \left(\sqrt{\frac{1}{2}} c F(t)^{a_2}\, G(t)^{b_2}\, e^{(c_2 x)}\right) dy + \left(-i\sqrt{\frac{1}{2}} c F(t)^{a_3}\, G(t)^{b_3}\, e^{(c_3 x)}\right) dz$$

Contravariant components

$$l = \left(-\frac{\sqrt{2}}{2\,c F(t)^{a_4}\, G(t)^{b_4}}\right) dt + \left(\frac{\sqrt{2}}{2\,c F(t)^{a_1}\, G(t)^{b_1}}\right) dx$$

$$n = \left(-\frac{\sqrt{2}}{2\,cF\left(t\right)^{a_4}\,G\left(t\right)^{b_4}}\right)\,dt + \left(-\frac{\sqrt{2}}{2\,cF\left(t\right)^{a_1}\,G\left(t\right)^{b_1}}\right)\,dx$$

$$m = \left(\frac{\sqrt{2}e^{(-c_2x)}}{2\,cF\left(t\right)^{a_2}\,G\left(t\right)^{b_2}}\right)\,dy + \left(\frac{i\,\sqrt{2}e^{(-c_3x)}}{2\,cF\left(t\right)^{a_3}\,G\left(t\right)^{b_3}}\right)\,dz$$

$$\overline{m} = \left(\frac{\sqrt{2}e^{(-c_2x)}}{2\,cF\left(t\right)^{a_2}\,G\left(t\right)^{b_2}}\right)\,dy + \left(-\frac{i\,\sqrt{2}e^{(-c_3x)}}{2\,cF\left(t\right)^{a_3}\,G\left(t\right)^{b_3}}\right)\,dz$$

Directional Derivatives

$$D = -\frac{\sqrt{2}}{2\,cF\left(t\right)^{a_4}\,G\left(t\right)^{b_4}}\,\partial_t + \frac{\sqrt{2}}{2\,cF\left(t\right)^{a_1}\,G\left(t\right)^{b_1}}\,\partial_x$$

$$\Delta = -\frac{\sqrt{2}}{2\,cF\left(t\right)^{a_4}\,G\left(t\right)^{b_4}}\,\partial_t - \frac{\sqrt{2}}{2\,cF\left(t\right)^{a_1}\,G\left(t\right)^{b_1}}\,\partial_x$$

$$\delta = \frac{\sqrt{2}e^{(-c_2x)}}{2\,cF\left(t\right)^{a_2}\,G\left(t\right)^{b_2}}\,\partial_y + \frac{i\,\sqrt{2}e^{(-c_3x)}}{2\,cF\left(t\right)^{a_3}\,G\left(t\right)^{b_3}}\,\partial_z$$

$$\overline{\delta} = \frac{\sqrt{2}e^{(-c_2x)}}{2\,cF\left(t\right)^{a_2}\,G\left(t\right)^{b_2}}\,\partial_y - \frac{i\,\sqrt{2}e^{(-c_3x)}}{2\,cF\left(t\right)^{a_3}\,G\left(t\right)^{b_3}}\,\partial_z$$

Spin Coefficients

$$\begin{aligned}\rho = &-\frac{\sqrt{2}c_2F\left(t\right)^{-a_1}\,F\left(t\right)^{-a_4}\,F\left(t\right)^{a_4}\,G\left(t\right)^{-b_1}\,G\left(t\right)^{-b_4}\,G\left(t\right)^{b_4}}{4\,c} - \frac{\sqrt{2}c_3F\left(t\right)^{-a_1}\,F\left(t\right)^{-a_4}\,F\left(t\right)^{a_4}\,G\left(t\right)^{-b_1}\,G\left(t\right)^{-b_4}\,G\left(t\right)^{b_4}}{4\,c}\\ &+\frac{\sqrt{2}a_2F\left(t\right)^{-a_1}\,F\left(t\right)^{a_1}\,F\left(t\right)^{-a_4}\,G\left(t\right)^{-b_1}\,G\left(t\right)^{b_1}\,G\left(t\right)^{-b_4}\,\frac{\partial}{\partial t}F\left(t\right)}{4\,cF\left(t\right)} + \frac{\sqrt{2}a_3F\left(t\right)^{-a_1}\,F\left(t\right)^{a_1}\,F\left(t\right)^{-a_4}\,G\left(t\right)^{-b_1}\,G\left(t\right)^{b_1}\,G\left(t\right)^{-b_4}\,\frac{\partial}{\partial t}F\left(t\right)}{4\,cF\left(t\right)}\\ &+\frac{\sqrt{2}b_2F\left(t\right)^{-a_1}\,F\left(t\right)^{a_1}\,F\left(t\right)^{-a_4}\,G\left(t\right)^{-b_1}\,G\left(t\right)^{b_1}\,G\left(t\right)^{-b_4}\,\frac{\partial}{\partial t}G\left(t\right)}{4\,cG\left(t\right)} + \frac{\sqrt{2}b_3F\left(t\right)^{-a_1}\,F\left(t\right)^{a_1}\,F\left(t\right)^{-a_4}\,G\left(t\right)^{-b_1}\,G\left(t\right)^{b_1}\,G\left(t\right)^{-b_4}\,\frac{\partial}{\partial t}G\left(t\right)}{4\,cG\left(t\right)}\end{aligned}$$

$$\sigma = -\frac{\sqrt{2}c_2 F(t)^{-a_1} F(t)^{-a_4} F(t)^{a_4} G(t)^{-b_1} G(t)^{-b_4} G(t)^{b_4}}{4\,c} + \frac{\sqrt{2}c_3 F(t)^{-a_1} F(t)^{-a_4} F(t)^{a_4} G(t)^{-b_1} G(t)^{-b_4} G(t)^{b_4}}{4\,c}$$
$$+ \frac{\sqrt{2}a_2 F(t)^{-a_1} F(t)^{a_1} F(t)^{-a_4} G(t)^{-b_1} G(t)^{b_1} G(t)^{-b_4} \frac{\partial}{\partial t}F(t)}{4\,cF(t)} - \frac{\sqrt{2}a_3 F(t)^{-a_1} F(t)^{a_1} F(t)^{-a_4} G(t)^{-b_1} G(t)^{b_1} G(t)^{-b_4} \frac{\partial}{\partial t}F(t)}{4\,cF(t)}$$
$$+ \frac{\sqrt{2}b_2 F(t)^{-a_1} F(t)^{a_1} F(t)^{-a_4} G(t)^{-b_1} G(t)^{b_1} G(t)^{-b_4} \frac{\partial}{\partial t}G(t)}{4\,cG(t)} - \frac{\sqrt{2}b_3 F(t)^{-a_1} F(t)^{a_1} F(t)^{-a_4} G(t)^{-b_1} G(t)^{b_1} G(t)^{-b_4} \frac{\partial}{\partial t}G(t)}{4\,cG(t)}$$

$$\kappa = 0$$

$$\mu = -\frac{\sqrt{2}c_2 F(t)^{-a_1} F(t)^{-a_4} F(t)^{a_4} G(t)^{-b_1} G(t)^{-b_4} G(t)^{b_4}}{4\,c} - \frac{\sqrt{2}c_3 F(t)^{-a_1} F(t)^{-a_4} F(t)^{a_4} G(t)^{-b_1} G(t)^{-b_4} G(t)^{b_4}}{4\,c}$$
$$- \frac{\sqrt{2}a_2 F(t)^{-a_1} F(t)^{a_1} F(t)^{-a_4} G(t)^{-b_1} G(t)^{b_1} G(t)^{-b_4} \frac{\partial}{\partial t}F(t)}{4\,cF(t)} - \frac{\sqrt{2}a_3 F(t)^{-a_1} F(t)^{a_1} F(t)^{-a_4} G(t)^{-b_1} G(t)^{b_1} G(t)^{-b_4} \frac{\partial}{\partial t}F(t)}{4\,cF(t)}$$
$$- \frac{\sqrt{2}b_2 F(t)^{-a_1} F(t)^{a_1} F(t)^{-a_4} G(t)^{-b_1} G(t)^{b_1} G(t)^{-b_4} \frac{\partial}{\partial t}G(t)}{4\,cG(t)} - \frac{\sqrt{2}b_3 F(t)^{-a_1} F(t)^{a_1} F(t)^{-a_4} G(t)^{-b_1} G(t)^{b_1} G(t)^{-b_4} \frac{\partial}{\partial t}G(t)}{4\,cG(t)}$$

$$\lambda = -\frac{\sqrt{2}c_2 F(t)^{-a_1} F(t)^{-a_4} F(t)^{a_4} G(t)^{-b_1} G(t)^{-b_4} G(t)^{b_4}}{4\,c} + \frac{\sqrt{2}c_3 F(t)^{-a_1} F(t)^{-a_4} F(t)^{a_4} G(t)^{-b_1} G(t)^{-b_4} G(t)^{b_4}}{4\,c}$$
$$- \frac{\sqrt{2}a_2 F(t)^{-a_1} F(t)^{a_1} F(t)^{-a_4} G(t)^{-b_1} G(t)^{b_1} G(t)^{-b_4} \frac{\partial}{\partial t}F(t)}{4\,cF(t)} + \frac{\sqrt{2}a_3 F(t)^{-a_1} F(t)^{a_1} F(t)^{-a_4} G(t)^{-b_1} G(t)^{b_1} G(t)^{-b_4} \frac{\partial}{\partial t}F(t)}{4\,cF(t)}$$
$$- \frac{\sqrt{2}b_2 F(t)^{-a_1} F(t)^{a_1} F(t)^{-a_4} G(t)^{-b_1} G(t)^{b_1} G(t)^{-b_4} \frac{\partial}{\partial t}G(t)}{4\,cG(t)} + \frac{\sqrt{2}b_3 F(t)^{-a_1} F(t)^{a_1} F(t)^{-a_4} G(t)^{-b_1} G(t)^{b_1} G(t)^{-b_4} \frac{\partial}{\partial t}G(t)}{4\,cG(t)}$$

$$\nu = 0$$

$$\alpha = 0$$

$$\beta = 0$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = -\frac{\sqrt{2}a_1 F(t)^{-a_4} G(t)^{-b_4} \frac{\partial}{\partial t} F(t)}{4\,cF(t)} - \frac{\sqrt{2}b_1 F(t)^{-a_4} G(t)^{-b_4} \frac{\partial}{\partial t} G(t)}{4\,cG(t)}$$

$$\gamma = \frac{\sqrt{2}a_1 F(t)^{-a_4} G(t)^{-b_4} \frac{\partial}{\partial t} F(t)}{4\,cF(t)} + \frac{\sqrt{2}b_1 F(t)^{-a_4} G(t)^{-b_4} \frac{\partial}{\partial t} G(t)}{4\,cG(t)}$$

Ricci Tensor Components

$$
\begin{aligned}
\Phi_{00} = & -\frac{a_1 c_2 F(t)^{-2a_1} F(t)^{a_1} F(t)^{-2a_4} F(t)^{a_4} G(t)^{-2b_1} G(t)^{b_1} G(t)^{-2b_4} G(t)^{b_4} \frac{\partial}{\partial t} F(t)}{2c^2 F(t)} + \frac{a_2 c_2 F(t)^{-2a_1} F(t)^{a_1} F(t)^{-2a_4} F(t)^{a_4} G(t)^{-2b_1} G(t)^{b_1} G(t)^{-2b_4} G(t)^{b_4} \frac{\partial}{\partial t} F(t)}{2c^2 F(t)} \\
& - \frac{a_1 c_3 F(t)^{-2a_1} F(t)^{a_1} F(t)^{-2a_4} F(t)^{a_4} G(t)^{-2b_1} G(t)^{b_1} G(t)^{-2b_4} G(t)^{b_4} \frac{\partial}{\partial t} F(t)}{2c^2 F(t)} + \frac{a_3 c_3 F(t)^{-2a_1} F(t)^{a_1} F(t)^{-2a_4} F(t)^{a_4} G(t)^{-2b_1} G(t)^{b_1} G(t)^{-2b_4} G(t)^{b_4} \frac{\partial}{\partial t} F(t)}{2c^2 F(t)} \\
& - \frac{b_1 c_2 F(t)^{-2a_1} F(t)^{a_1} F(t)^{-2a_4} F(t)^{a_4} G(t)^{-2b_1} G(t)^{b_1} G(t)^{-2b_4} G(t)^{b_4} \frac{\partial}{\partial t} G(t)}{2c^2 G(t)} + \frac{b_2 c_2 F(t)^{-2a_1} F(t)^{a_1} F(t)^{-2a_4} F(t)^{a_4} G(t)^{-2b_1} G(t)^{b_1} G(t)^{-2b_4} G(t)^{b_4} \frac{\partial}{\partial t} G(t)}{2c^2 G(t)} \\
& - \frac{b_1 c_3 F(t)^{-2a_1} F(t)^{a_1} F(t)^{-2a_4} F(t)^{a_4} G(t)^{-2b_1} G(t)^{b_1} G(t)^{-2b_4} G(t)^{b_4} \frac{\partial}{\partial t} G(t)}{2c^2 G(t)} + \frac{b_3 c_3 F(t)^{-2a_1} F(t)^{a_1} F(t)^{-2a_4} F(t)^{a_4} G(t)^{-2b_1} G(t)^{b_1} G(t)^{-2b_4} G(t)^{b_4} \frac{\partial}{\partial t} G(t)}{2c^2 G(t)} \\
& - \frac{{c_2}^2 F(t)^{-2a_1} F(t)^{2a_4} F(t)^{-2a_4} G(t)^{-2b_1} G(t)^{2b_4} G(t)^{-2b_4}}{4c^2} - \frac{{c_3}^2 F(t)^{-2a_1} F(t)^{2a_4} F(t)^{-2a_4} G(t)^{-2b_1} G(t)^{2b_4} G(t)^{-2b_4}}{4c^2} \\
& + \frac{a_1 a_2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{4c^2 F(t)^2} - \frac{{a_2}^2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{4c^2 F(t)^2} \\
& + \frac{a_1 a_3 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{4c^2 F(t)^2} - \frac{{a_3}^2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{4c^2 F(t)^2} \\
& + \frac{a_2 a_4 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{4c^2 F(t)^2} + \frac{a_3 a_4 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{4c^2 F(t)^2} \\
& + \frac{a_2 b_1 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{4c^2 F(t) G(t)} + \frac{a_3 b_1 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{4c^2 F(t) G(t)} \\
& + \frac{a_1 b_2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{4c^2 F(t) G(t)} - \frac{a_2 b_2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{2c^2 F(t) G(t)} \\
& + \frac{a_4 b_2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{4c^2 F(t) G(t)} + \frac{a_1 b_3 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{4c^2 F(t) G(t)} \\
& - \frac{a_3 b_3 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{2c^2 F(t) G(t)} + \frac{a_4 b_3 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{4c^2 F(t) G(t)} \\
& + \frac{a_2 b_4 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{4c^2 F(t) G(t)} + \frac{a_3 b_4 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{4c^2 F(t) G(t)} \\
& + \frac{b_1 b_2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2}{4c^2 G(t)^2} - \frac{{b_2}^2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2}{4c^2 G(t)^2} \\
& + \frac{b_1 b_3 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2}{4c^2 G(t)^2} - \frac{{b_3}^2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2}{4c^2 G(t)^2} \\
& + \frac{b_2 b_4 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2}{4c^2 G(t)^2} + \frac{b_3 b_4 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2}{4c^2 G(t)^2} \\
& + \frac{a_2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{4c^2 F(t)^2} + \frac{a_3 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{4c^2 F(t)^2} \\
& - \frac{a_2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial^2}{(\partial t)^2} F(t)}{4c^2 F(t)} - \frac{a_3 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial^2}{(\partial t)^2} F(t)}{4c^2 F(t)} \\
& \frac{b_2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2}{5} \quad b_2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2
\end{aligned}
$$

$\Phi_{01} = 0$

$$
\begin{aligned}
\Phi_{02} = & -\frac{{c_2}^2 F(t)^{-2a_1} F(t)^{2a_4} F(t)^{-2a_4} G(t)^{-2b_1} G(t)^{2b_4} G(t)^{-2b_4}}{4c^2} + \frac{{c_3}^2 F(t)^{-2a_1} F(t)^{2a_4} F(t)^{-2a_4} G(t)^{-2b_1} G(t)^{2b_4} G(t)^{-2b_4}}{4c^2} \\
& + \frac{a_1 a_2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{4c^2 F(t)^2} + \frac{{a_2}^2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{4c^2 F(t)^2} \\
& - \frac{a_1 a_3 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{4c^2 F(t)^2} - \frac{{a_3}^2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{4c^2 F(t)^2} \\
& - \frac{a_2 a_4 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{4c^2 F(t)^2} + \frac{a_3 a_4 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{4c^2 F(t)^2} \\
& + \frac{a_2 b_1 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{4c^2 F(t) G(t)} - \frac{a_3 b_1 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{4c^2 F(t) G(t)} \\
& + \frac{a_1 b_2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{4c^2 F(t) G(t)} + \frac{a_2 b_2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{2c^2 F(t) G(t)} \\
& - \frac{a_4 b_2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{4c^2 F(t) G(t)} - \frac{a_1 b_3 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{4c^2 F(t) G(t)} \\
& - \frac{a_3 b_3 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{2c^2 F(t) G(t)} + \frac{a_4 b_3 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{4c^2 F(t) G(t)} \\
& - \frac{a_2 b_4 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{4c^2 F(t) G(t)} + \frac{a_3 b_4 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{4c^2 F(t) G(t)} \\
& + \frac{b_1 b_2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2}{4c^2 G(t)^2} + \frac{{b_2}^2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2}{4c^2 G(t)^2} \\
& - \frac{b_1 b_3 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2}{4c^2 G(t)^2} - \frac{{b_3}^2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2}{4c^2 G(t)^2} \\
& - \frac{b_2 b_4 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2}{4c^2 G(t)^2} + \frac{b_3 b_4 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2}{4c^2 G(t)^2} \\
& - \frac{a_2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{4c^2 F(t)^2} + \frac{a_3 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{4c^2 F(t)^2} \\
& + \frac{a_2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial^2}{(\partial t)^2} F(t)}{4c^2 F(t)} - \frac{a_3 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial^2}{(\partial t)^2} F(t)}{4c^2 F(t)} \\
& - \frac{b_2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2}{4c^2 G(t)^2} + \frac{b_3 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2}{4c^2 G(t)^2} \\
& + \frac{b_2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial^2}{(\partial t)^2} G(t)}{4c^2 G(t)} - \frac{b_3 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial^2}{(\partial t)^2} G(t)}{4c^2 G(t)}
\end{aligned}
$$

$$\Phi_{10} = 0$$

$$\begin{aligned}
\Phi_{11} = & -\frac{c_2 c_3 F(t)^{-2a_1} F(t)^{2a_4} F(t)^{-2a_4} G(t)^{-2b_1} G(t)^{2b_4} G(t)^{-2b_4}}{4c^2} - \frac{{a_1}^2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{4c^2 F(t)^2} \\
& + \frac{a_2 a_3 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{4c^2 F(t)^2} + \frac{a_1 a_4 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{4c^2 F(t)^2} \\
& - \frac{a_1 b_1 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{2c^2 F(t) G(t)} + \frac{a_4 b_1 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{4c^2 F(t) G(t)} \\
& + \frac{a_3 b_2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{4c^2 F(t) G(t)} + \frac{a_2 b_3 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{4c^2 F(t) G(t)} \\
& + \frac{a_1 b_4 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{4c^2 F(t) G(t)} - \frac{{b_1}^2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2}{4c^2 G(t)^2} \\
& + \frac{b_2 b_3 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2}{4c^2 G(t)^2} + \frac{b_1 b_4 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2}{4c^2 G(t)^2} \\
& + \frac{a_1 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{4c^2 F(t)^2} - \frac{a_1 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial^2}{(\partial t)^2} F(t)}{4c^2 F(t)} \\
& + \frac{b_1 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2}{4c^2 G(t)^2} - \frac{b_1 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial^2}{(\partial t)^2} G(t)}{4c^2 G(t)}
\end{aligned}$$

$$\Phi_{12} = 0$$

$$
\begin{aligned}
\Phi_{20} = & -\frac{{c_2}^2 F(t)^{-2a_1} F(t)^{2a_4} F(t)^{-2a_4} G(t)^{-2b_1} G(t)^{2b_4} G(t)^{-2b_4}}{4c^2} + \frac{{c_3}^2 F(t)^{-2a_1} F(t)^{2a_4} F(t)^{-2a_4} G(t)^{-2b_1} G(t)^{2b_4} G(t)^{-2b_4}}{4c^2} \\
& + \frac{a_1 a_2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{4c^2 F(t)^2} + \frac{{a_2}^2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{4c^2 F(t)^2} \\
& - \frac{a_1 a_3 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{4c^2 F(t)^2} - \frac{{a_3}^2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{4c^2 F(t)^2} \\
& - \frac{a_2 a_4 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{4c^2 F(t)^2} + \frac{a_3 a_4 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{4c^2 F(t)^2} \\
& + \frac{a_2 b_1 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{4c^2 F(t) G(t)} - \frac{a_3 b_1 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{4c^2 F(t) G(t)} \\
& + \frac{a_1 b_2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{4c^2 F(t) G(t)} + \frac{a_2 b_2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{2c^2 F(t) G(t)} \\
& - \frac{a_4 b_2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{4c^2 F(t) G(t)} - \frac{a_1 b_3 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{4c^2 F(t) G(t)} \\
& - \frac{a_3 b_3 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{2c^2 F(t) G(t)} + \frac{a_4 b_3 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{4c^2 F(t) G(t)} \\
& - \frac{a_2 b_4 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{4c^2 F(t) G(t)} + \frac{a_3 b_4 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{4c^2 F(t) G(t)} \\
& + \frac{b_1 b_2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2}{4c^2 G(t)^2} + \frac{{b_2}^2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2}{4c^2 G(t)^2} \\
& - \frac{b_1 b_3 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2}{4c^2 G(t)^2} - \frac{{b_3}^2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2}{4c^2 G(t)^2} \\
& - \frac{b_2 b_4 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2}{4c^2 G(t)^2} + \frac{b_3 b_4 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2}{4c^2 G(t)^2} \\
& - \frac{a_2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{4c^2 F(t)^2} + \frac{a_3 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{4c^2 F(t)^2} \\
& + \frac{a_2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial^2}{(\partial t)^2} F(t)}{4c^2 F(t)} - \frac{a_3 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial^2}{(\partial t)^2} F(t)}{4c^2 F(t)} \\
& - \frac{b_2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2}{4c^2 G(t)^2} + \frac{b_3 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2}{4c^2 G(t)^2} \\
& + \frac{b_2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial^2}{(\partial t)^2} G(t)}{4c^2 G(t)} - \frac{b_3 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial^2}{(\partial t)^2} G(t)}{4c^2 G(t)}
\end{aligned}
$$

$\Phi_{21} = 0$

$$
\begin{aligned}
\Phi_{22} = {} & \frac{a_1 c_2 F(t)^{-2a_1} F(t)^{a_1} F(t)^{-2a_4} F(t)^{a_4} G(t)^{-2b_1} G(t)^{b_1} G(t)^{-2b_4} G(t)^{b_4} \frac{\partial}{\partial t} F(t)}{2c^2 F(t)} - \frac{a_2 c_2 F(t)^{-2a_1} F(t)^{a_1} F(t)^{-2a_4} F(t)^{a_4} G(t)^{-2b_1} G(t)^{b_1} G(t)^{-2b_4} G(t)^{b_4} \frac{\partial}{\partial t} F(t)}{2c^2 F(t)} \\
& + \frac{a_1 c_3 F(t)^{-2a_1} F(t)^{a_1} F(t)^{-2a_4} F(t)^{a_4} G(t)^{-2b_1} G(t)^{b_1} G(t)^{-2b_4} G(t)^{b_4} \frac{\partial}{\partial t} F(t)}{2c^2 F(t)} - \frac{a_3 c_3 F(t)^{-2a_1} F(t)^{a_1} F(t)^{-2a_4} F(t)^{a_4} G(t)^{-2b_1} G(t)^{b_1} G(t)^{-2b_4} G(t)^{b_4} \frac{\partial}{\partial t} F(t)}{2c^2 F(t)} \\
& + \frac{b_1 c_2 F(t)^{-2a_1} F(t)^{a_1} F(t)^{-2a_4} F(t)^{a_4} G(t)^{-2b_1} G(t)^{b_1} G(t)^{-2b_4} G(t)^{b_4} \frac{\partial}{\partial t} G(t)}{2c^2 G(t)} - \frac{b_2 c_2 F(t)^{-2a_1} F(t)^{a_1} F(t)^{-2a_4} F(t)^{a_4} G(t)^{-2b_1} G(t)^{b_1} G(t)^{-2b_4} G(t)^{b_4} \frac{\partial}{\partial t} G(t)}{2c^2 G(t)} \\
& + \frac{b_1 c_3 F(t)^{-2a_1} F(t)^{a_1} F(t)^{-2a_4} F(t)^{a_4} G(t)^{-2b_1} G(t)^{b_1} G(t)^{-2b_4} G(t)^{b_4} \frac{\partial}{\partial t} G(t)}{2c^2 G(t)} - \frac{b_3 c_3 F(t)^{-2a_1} F(t)^{a_1} F(t)^{-2a_4} F(t)^{a_4} G(t)^{-2b_1} G(t)^{b_1} G(t)^{-2b_4} G(t)^{b_4} \frac{\partial}{\partial t} G(t)}{2c^2 G(t)} \\
& - \frac{{c_2}^2 F(t)^{-2a_1} F(t)^{2a_4} F(t)^{-2a_4} G(t)^{-2b_1} G(t)^{2b_4} G(t)^{-2b_4}}{4c^2} - \frac{{c_3}^2 F(t)^{-2a_1} F(t)^{2a_4} F(t)^{-2a_4} G(t)^{-2b_1} G(t)^{2b_4} G(t)^{-2b_4}}{4c^2} \\
& + \frac{a_1 a_2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{4c^2 F(t)^2} - \frac{{a_2}^2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{4c^2 F(t)^2} \\
& + \frac{a_1 a_3 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{4c^2 F(t)^2} - \frac{{a_3}^2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{4c^2 F(t)^2} \\
& + \frac{a_2 a_4 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{4c^2 F(t)^2} + \frac{a_3 a_4 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{4c^2 F(t)^2} \\
& + \frac{a_2 b_1 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{4c^2 F(t) G(t)} + \frac{a_3 b_1 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{4c^2 F(t) G(t)} \\
& + \frac{a_1 b_2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{4c^2 F(t) G(t)} - \frac{a_2 b_2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{2c^2 F(t) G(t)} \\
& + \frac{a_4 b_2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{4c^2 F(t) G(t)} + \frac{a_1 b_3 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{4c^2 F(t) G(t)} \\
& - \frac{a_3 b_3 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{2c^2 F(t) G(t)} + \frac{a_4 b_3 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{4c^2 F(t) G(t)} \\
& + \frac{a_2 b_4 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{4c^2 F(t) G(t)} + \frac{a_3 b_4 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{4c^2 F(t) G(t)} \\
& + \frac{b_1 b_2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2}{4c^2 G(t)^2} - \frac{{b_2}^2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2}{4c^2 G(t)^2} \\
& + \frac{b_1 b_3 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2}{4c^2 G(t)^2} - \frac{{b_3}^2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2}{4c^2 G(t)^2} \\
& + \frac{b_2 b_4 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2}{4c^2 G(t)^2} + \frac{b_3 b_4 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2}{4c^2 G(t)^2} \\
& + \frac{a_2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{4c^2 F(t)^2} + \frac{a_3 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{4c^2 F(t)^2} \\
& - \frac{a_2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial^2}{(\partial t)^2} F(t)}{4c^2 F(t)} - \frac{a_3 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial^2}{(\partial t)^2} F(t)}{4c^2 F(t)} \\
& \qquad b_2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2 \qquad b_2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2
\end{aligned}
$$

$$
\begin{aligned}
\Lambda = & -\frac{{c_2}^2 F(t)^{-2a_1} F(t)^{2a_4} F(t)^{-2a_4} G(t)^{-2b_1} G(t)^{2b_4} G(t)^{-2b_4}}{12c^2} - \frac{c_2 c_3 F(t)^{-2a_1} F(t)^{2a_4} F(t)^{-2a_4} G(t)^{-2b_1} G(t)^{2b_4} G(t)^{-2b_4}}{12c^2} \\
& - \frac{{c_3}^2 F(t)^{-2a_1} F(t)^{2a_4} F(t)^{-2a_4} G(t)^{-2b_1} G(t)^{2b_4} G(t)^{-2b_4}}{12c^2} + \frac{{a_1}^2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{12c^2 F(t)^2} \\
& + \frac{a_1 a_2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{12c^2 F(t)^2} + \frac{{a_2}^2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{12c^2 F(t)^2} \\
& + \frac{a_1 a_3 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{12c^2 F(t)^2} + \frac{a_2 a_3 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{12c^2 F(t)^2} \\
& + \frac{{a_3}^2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{12c^2 F(t)^2} - \frac{a_1 a_4 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{12c^2 F(t)^2} \\
& - \frac{a_2 a_4 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{12c^2 F(t)^2} - \frac{a_3 a_4 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{12c^2 F(t)^2} \\
& + \frac{a_1 b_1 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{6c^2 F(t) G(t)} + \frac{a_2 b_1 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{12c^2 F(t) G(t)} \\
& + \frac{a_3 b_1 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{12c^2 F(t) G(t)} - \frac{a_4 b_1 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{12c^2 F(t) G(t)} \\
& + \frac{a_1 b_2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{12c^2 F(t) G(t)} + \frac{a_2 b_2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{6c^2 F(t) G(t)} \\
& + \frac{a_3 b_2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{12c^2 F(t) G(t)} - \frac{a_4 b_2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{12c^2 F(t) G(t)} \\
& + \frac{a_1 b_3 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{12c^2 F(t) G(t)} + \frac{a_2 b_3 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{12c^2 F(t) G(t)} \\
& + \frac{a_3 b_3 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{6c^2 F(t) G(t)} - \frac{a_4 b_3 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{12c^2 F(t) G(t)} \\
& - \frac{a_1 b_4 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{12c^2 F(t) G(t)} - \frac{a_2 b_4 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{12c^2 F(t) G(t)} \\
& - \frac{a_3 b_4 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{12c^2 F(t) G(t)} + \frac{{b_1}^2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2}{12c^2 G(t)^2} \\
& + \frac{b_1 b_2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2}{12c^2 G(t)^2} + \frac{{b_2}^2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2}{12c^2 G(t)^2} \\
& + \frac{b_1 b_3 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2}{12c^2 G(t)^2} + \frac{b_2 b_3 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2}{12c^2 G(t)^2} \\
& + \frac{{b_3}^2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2}{12c^2 G(t)^2} - \frac{b_1 b_4 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2}{12c^2 G(t)^2} \\
& - \frac{b_2 b_4 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2}{12c^2 G(t)^2} - \frac{b_3 b_4 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2}{12c^2 G(t)^2} \\
& - \frac{a_1 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{12} - \frac{a_2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{12}
\end{aligned}
$$

Weyl Tensor Components

$$
\begin{aligned}
\Psi_0 = & -\frac{a_1 c_2 F(t)^{-2a_1} F(t)^{a_1} F(t)^{-2a_4} F(t)^{a_4} G(t)^{-2b_1} G(t)^{b_1} G(t)^{-2b_4} G(t)^{b_4} \frac{\partial}{\partial t} F(t)}{2c^2 F(t)} + \frac{a_2 c_2 F(t)^{-2a_1} F(t)^{a_1} F(t)^{-2a_4} F(t)^{a_4} G(t)^{-2b_1} G(t)^{b_1} G(t)^{-2b_4} G(t)^{b_4} \frac{\partial}{\partial t} F(t)}{2c^2 F(t)} \\
& + \frac{a_1 c_3 F(t)^{-2a_1} F(t)^{a_1} F(t)^{-2a_4} F(t)^{a_4} G(t)^{-2b_1} G(t)^{b_1} G(t)^{-2b_4} G(t)^{b_4} \frac{\partial}{\partial t} F(t)}{2c^2 F(t)} - \frac{a_3 c_3 F(t)^{-2a_1} F(t)^{a_1} F(t)^{-2a_4} F(t)^{a_4} G(t)^{-2b_1} G(t)^{b_1} G(t)^{-2b_4} G(t)^{b_4} \frac{\partial}{\partial t} F(t)}{2c^2 F(t)} \\
& - \frac{b_1 c_2 F(t)^{-2a_1} F(t)^{a_1} F(t)^{-2a_4} F(t)^{a_4} G(t)^{-2b_1} G(t)^{b_1} G(t)^{-2b_4} G(t)^{b_4} \frac{\partial}{\partial t} G(t)}{2c^2 G(t)} + \frac{b_2 c_2 F(t)^{-2a_1} F(t)^{a_1} F(t)^{-2a_4} F(t)^{a_4} G(t)^{-2b_1} G(t)^{b_1} G(t)^{-2b_4} G(t)^{b_4} \frac{\partial}{\partial t} G(t)}{2c^2 G(t)} \\
& + \frac{b_1 c_3 F(t)^{-2a_1} F(t)^{a_1} F(t)^{-2a_4} F(t)^{a_4} G(t)^{-2b_1} G(t)^{b_1} G(t)^{-2b_4} G(t)^{b_4} \frac{\partial}{\partial t} G(t)}{2c^2 G(t)} - \frac{b_3 c_3 F(t)^{-2a_1} F(t)^{a_1} F(t)^{-2a_4} F(t)^{a_4} G(t)^{-2b_1} G(t)^{b_1} G(t)^{-2b_4} G(t)^{b_4} \frac{\partial}{\partial t} G(t)}{2c^2 G(t)} \\
& - \frac{{c_2}^2 F(t)^{-2a_1} F(t)^{2a_4} F(t)^{-2a_4} G(t)^{-2b_1} G(t)^{2b_4} G(t)^{-2b_4}}{4c^2} + \frac{{c_3}^2 F(t)^{-2a_1} F(t)^{2a_4} F(t)^{-2a_4} G(t)^{-2b_1} G(t)^{2b_4} G(t)^{-2b_4}}{4c^2} \\
& + \frac{a_1 a_2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{4c^2 F(t)^2} - \frac{{a_2}^2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{4c^2 F(t)^2} \\
& - \frac{a_1 a_3 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{4c^2 F(t)^2} + \frac{{a_3}^2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{4c^2 F(t)^2} \\
& + \frac{a_2 a_4 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{4c^2 F(t)^2} - \frac{a_3 a_4 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{4c^2 F(t)^2} \\
& + \frac{a_2 b_1 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{4c^2 F(t) G(t)} - \frac{a_3 b_1 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{4c^2 F(t) G(t)} \\
& + \frac{a_1 b_2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{4c^2 F(t) G(t)} - \frac{a_2 b_2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{2c^2 F(t) G(t)} \\
& + \frac{a_4 b_2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{4c^2 F(t) G(t)} - \frac{a_1 b_3 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{4c^2 F(t) G(t)} \\
& + \frac{a_3 b_3 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{2c^2 F(t) G(t)} - \frac{a_4 b_3 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{4c^2 F(t) G(t)} \\
& + \frac{a_2 b_4 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{4c^2 F(t) G(t)} - \frac{a_3 b_4 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{4c^2 F(t) G(t)} \\
& + \frac{b_1 b_2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2}{4c^2 G(t)^2} - \frac{{b_2}^2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2}{4c^2 G(t)^2} \\
& - \frac{b_1 b_3 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2}{4c^2 G(t)^2} + \frac{{b_3}^2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2}{4c^2 G(t)^2} \\
& + \frac{b_2 b_4 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2}{4c^2 G(t)^2} - \frac{b_3 b_4 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2}{4c^2 G(t)^2} \\
& + \frac{a_2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{4c^2 F(t)^2} - \frac{a_3 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{4c^2 F(t)^2} \\
& - \frac{a_2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial^2}{(\partial t)^2} F(t)}{4c^2 F(t)} + \frac{a_3 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial^2}{(\partial t)^2} F(t)}{4c^2 F(t)} \\
& \quad b_2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2 \quad b_2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2
\end{aligned}
$$

$\Psi_1 = 0$

$$
\begin{aligned}
\Psi_2 = & -\frac{{c_2}^2 F(t)^{-2a_1} F(t)^{2a_4} F(t)^{-2a_4} G(t)^{-2b_1} G(t)^{2b_4} G(t)^{-2b_4}}{12c^2} + \frac{c_2 c_3 F(t)^{-2a_1} F(t)^{2a_4} F(t)^{-2a_4} G(t)^{-2b_1} G(t)^{2b_4} G(t)^{-2b_4}}{6c^2} \\
& - \frac{{c_3}^2 F(t)^{-2a_1} F(t)^{2a_4} F(t)^{-2a_4} G(t)^{-2b_1} G(t)^{2b_4} G(t)^{-2b_4}}{12c^2} - \frac{{a_1}^2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{6c^2 F(t)^2} \\
& + \frac{a_1 a_2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{12c^2 F(t)^2} + \frac{{a_2}^2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{12c^2 F(t)^2} \\
& + \frac{a_1 a_3 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{12c^2 F(t)^2} - \frac{a_2 a_3 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{6c^2 F(t)^2} \\
& + \frac{{a_3}^2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{12c^2 F(t)^2} + \frac{a_1 a_4 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{6c^2 F(t)^2} \\
& - \frac{a_2 a_4 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{12c^2 F(t)^2} - \frac{a_3 a_4 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{12c^2 F(t)^2} \\
& - \frac{a_1 b_1 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{3c^2 F(t) G(t)} + \frac{a_2 b_1 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{12c^2 F(t) G(t)} \\
& + \frac{a_3 b_1 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{12c^2 F(t) G(t)} + \frac{a_4 b_1 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{6c^2 F(t) G(t)} \\
& + \frac{a_1 b_2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{12c^2 F(t) G(t)} + \frac{a_2 b_2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{6c^2 F(t) G(t)} \\
& - \frac{a_3 b_2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{6c^2 F(t) G(t)} - \frac{a_4 b_2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{12c^2 F(t) G(t)} \\
& + \frac{a_1 b_3 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{12c^2 F(t) G(t)} - \frac{a_2 b_3 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{6c^2 F(t) G(t)} \\
& + \frac{a_3 b_3 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{6c^2 F(t) G(t)} - \frac{a_4 b_3 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{12c^2 F(t) G(t)} \\
& + \frac{a_1 b_4 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{6c^2 F(t) G(t)} - \frac{a_2 b_4 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{12c^2 F(t) G(t)} \\
& - \frac{a_3 b_4 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{12c^2 F(t) G(t)} - \frac{{b_1}^2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2}{6c^2 G(t)^2} \\
& + \frac{b_1 b_2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2}{12c^2 G(t)^2} + \frac{{b_2}^2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2}{12c^2 G(t)^2} \\
& + \frac{b_1 b_3 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2}{12c^2 G(t)^2} - \frac{b_2 b_3 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2}{6c^2 G(t)^2} \\
& + \frac{{b_3}^2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2}{12c^2 G(t)^2} + \frac{b_1 b_4 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2}{6c^2 G(t)^2} \\
& - \frac{b_2 b_4 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2}{12c^2 G(t)^2} - \frac{b_3 b_4 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2}{12c^2 G(t)^2}
\end{aligned}
$$

$\Psi_3 = 0$

$$
\begin{aligned}
\Psi_4 = {} & \frac{a_1 c_2 F(t)^{-2a_1} F(t)^{a_1} F(t)^{-2a_4} F(t)^{a_4} G(t)^{-2b_1} G(t)^{b_1} G(t)^{-2b_4} G(t)^{b_4} \frac{\partial}{\partial t} F(t)}{2c^2 F(t)} - \frac{a_2 c_2 F(t)^{-2a_1} F(t)^{a_1} F(t)^{-2a_4} F(t)^{a_4} G(t)^{-2b_1} G(t)^{b_1} G(t)^{-2b_4} G(t)^{b_4} \frac{\partial}{\partial t} F(t)}{2c^2 F(t)} \\
& - \frac{a_1 c_3 F(t)^{-2a_1} F(t)^{a_1} F(t)^{-2a_4} F(t)^{a_4} G(t)^{-2b_1} G(t)^{b_1} G(t)^{-2b_4} G(t)^{b_4} \frac{\partial}{\partial t} F(t)}{2c^2 F(t)} + \frac{a_3 c_3 F(t)^{-2a_1} F(t)^{a_1} F(t)^{-2a_4} F(t)^{a_4} G(t)^{-2b_1} G(t)^{b_1} G(t)^{-2b_4} G(t)^{b_4} \frac{\partial}{\partial t} F(t)}{2c^2 F(t)} \\
& + \frac{b_1 c_2 F(t)^{-2a_1} F(t)^{a_1} F(t)^{-2a_4} F(t)^{a_4} G(t)^{-2b_1} G(t)^{b_1} G(t)^{-2b_4} G(t)^{b_4} \frac{\partial}{\partial t} G(t)}{2c^2 G(t)} - \frac{b_2 c_2 F(t)^{-2a_1} F(t)^{a_1} F(t)^{-2a_4} F(t)^{a_4} G(t)^{-2b_1} G(t)^{b_1} G(t)^{-2b_4} G(t)^{b_4} \frac{\partial}{\partial t} G(t)}{2c^2 G(t)} \\
& - \frac{b_1 c_3 F(t)^{-2a_1} F(t)^{a_1} F(t)^{-2a_4} F(t)^{a_4} G(t)^{-2b_1} G(t)^{b_1} G(t)^{-2b_4} G(t)^{b_4} \frac{\partial}{\partial t} G(t)}{2c^2 G(t)} + \frac{b_3 c_3 F(t)^{-2a_1} F(t)^{a_1} F(t)^{-2a_4} F(t)^{a_4} G(t)^{-2b_1} G(t)^{b_1} G(t)^{-2b_4} G(t)^{b_4} \frac{\partial}{\partial t} G(t)}{2c^2 G(t)} \\
& - \frac{{c_2}^2 F(t)^{-2a_1} F(t)^{2a_4} F(t)^{-2a_4} G(t)^{-2b_1} G(t)^{2b_4} G(t)^{-2b_4}}{4c^2} + \frac{{c_3}^2 F(t)^{-2a_1} F(t)^{2a_4} F(t)^{-2a_4} G(t)^{-2b_1} G(t)^{2b_4} G(t)^{-2b_4}}{4c^2} \\
& + \frac{a_1 a_2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{4c^2 F(t)^2} - \frac{{a_2}^2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{4c^2 F(t)^2} \\
& - \frac{a_1 a_3 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{4c^2 F(t)^2} + \frac{{a_3}^2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{4c^2 F(t)^2} \\
& + \frac{a_2 a_4 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{4c^2 F(t)^2} - \frac{a_3 a_4 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{4c^2 F(t)^2} \\
& + \frac{a_2 b_1 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{4c^2 F(t) G(t)} - \frac{a_3 b_1 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{4c^2 F(t) G(t)} \\
& + \frac{a_1 b_2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{4c^2 F(t) G(t)} - \frac{a_2 b_2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{2c^2 F(t) G(t)} \\
& + \frac{a_4 b_2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{4c^2 F(t) G(t)} - \frac{a_1 b_3 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{4c^2 F(t) G(t)} \\
& + \frac{a_3 b_3 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{2c^2 F(t) G(t)} - \frac{a_4 b_3 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{4c^2 F(t) G(t)} \\
& + \frac{a_2 b_4 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{4c^2 F(t) G(t)} - \frac{a_3 b_4 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t) \frac{\partial}{\partial t} G(t)}{4c^2 F(t) G(t)} \\
& + \frac{b_1 b_2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2}{4c^2 G(t)^2} - \frac{{b_2}^2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2}{4c^2 G(t)^2} \\
& - \frac{b_1 b_3 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2}{4c^2 G(t)^2} + \frac{{b_3}^2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2}{4c^2 G(t)^2} \\
& + \frac{b_2 b_4 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2}{4c^2 G(t)^2} - \frac{b_3 b_4 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2}{4c^2 G(t)^2} \\
& + \frac{a_2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{4c^2 F(t)^2} - \frac{a_3 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} F(t)^2}{4c^2 F(t)^2} \\
& - \frac{a_2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial^2}{(\partial t)^2} F(t)}{4c^2 F(t)} + \frac{a_3 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial^2}{(\partial t)^2} F(t)}{4c^2 F(t)} \\
& \frac{b_2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2}{18} - \frac{b_2 F(t)^{2a_1} F(t)^{-2a_1} F(t)^{-2a_4} G(t)^{2b_1} G(t)^{-2b_1} G(t)^{-2b_4} \frac{\partial}{\partial t} G(t)^2}{}
\end{aligned}
$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "I"